1 Motivation for complex numbers: the equation $x^2 = -1$

The field of real numbers $\mathbb{R}$ has many desirable properties: in particular, it satisfies the field and order axioms and the completeness axiom. However, the ordered-field structure also imposes an important limitation. For every $x \in \mathbb{R}$,

$$x^2 \geq 0.$$

We proved that this is true in every ordered field (see HW4). Consequently, an equation of the form $x^2 = -a$ has no solution in $\mathbb{R}$ whenever $a > 0$. In particular, since $-1 < 0$, $x^2 = -1$ has no real solution.

Suppose we had an element i satisfying $i^2 = -1$. Then we can solve any equation $x^2 = -a, a > 0$, by $x = \pm i\sqrt{a}$, since $x^2 = (\pm i\sqrt{a})^2 = i^2 a = -a$.

More importantly, once we have i with $i^2 = -1$, we can solve every quadratic equation, including those that have no real roots.

Consider

$$ax^2 + bx + c = 0, \qquad a \neq 0.$$

Dividing by a gives

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

Completing the square,

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}. \tag{1}$$

So if we can solve an equation of the form $y^2 = F$ for $F \in \mathbb{R}$, then we can solve the general quadratic equation. We call $D = b^2 - 4ac$ the discriminant, and if it is nonnegative we get the solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

If $D = b^2 - 4ac < 0$, we can solve (1) via

$$\boxed{x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}}.$$

So if we can work with expressions $A + iB$, where $A, B \in \mathbb{R}$, we can solve every quadratic equation with real coefficients. In other words we need to define expressions of the form $A + iB$ and be able to do operations with them.

Our goal is therefore to construct the smallest possible field containing $\mathbb{R}$ together with an element i satisfying $i^2 = -1$ so we need to define operations between expressions of the form $x + iy$, where $x, y \in \mathbb{R}$.

2 Interlude: polynomial equations and number systems

One way to understand why we keep enlarging our number systems is to ask the following question: If a polynomial has coefficients in a given number system, must its solutions also belong to that number system?
For the number systems we have encountered so far, the answer is generally no.
For example, consider

$$x + 2 = 0.$$

This is a polynomial equation whose coefficients 1 and 2 lie in $\mathbb{N}$. However, its solution is

$$x = -2,$$

which does not belong to $\mathbb{N}$. To obtain a solution, we need to enlarge our number system from $\mathbb{N}$ to $\mathbb{Z}$.
But passing to $\mathbb{Z}$ does not solve every polynomial equation with coefficients in $\mathbb{Z}$. For example,

$$2x + 1 = 0$$

has coefficients in $\mathbb{Z}$, but its solution

$$x = -\frac{1}{2}$$

does not lie in $\mathbb{Z}$. It does, however, lie in $\mathbb{Q}$.
Again, $\mathbb{Q}$ is still not enough. The equation

$$x^2 - 2 = 0$$

has coefficients in $\mathbb{Q}$, but the solutions

$$x = \pm\sqrt{2}$$

do not belong to $\mathbb{Q}$. They do belong to $\mathbb{R}$.
Finally, even $\mathbb{R}$ does not contain solutions to every polynomial equation with coefficients in $\mathbb{R}$. The equation

$$x^2 + 1 = 0$$

has coefficients in $\mathbb{R}$, but solving

$$x^2 + 1 = 0$$

requires

$$x^2 = -1,$$

which has no solution in $\mathbb{R}$. After constructing $\mathbb{C}$, however, we obtain the solutions

$$x = \pm i.$$

Thus we have the chain

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C},$$

where at each stage we can find a polynomial whose coefficients belong to one number system but whose roots require us to pass to a larger one.
This naturally raises the question: does the process continue?
Do we need to construct some still larger number system containing $\mathbb{C}$ in order to solve polynomial equations whose coefficients are complex numbers?
Remarkably, the answer is no.
The **Fundamental Theorem of Algebra** says that every nonconstant polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.
In fact, a polynomial of degree n with complex coefficients has exactly n complex roots, counted with multiplicity. So, from the point of view of solving polynomial equations, $\mathbb{C}$ is where this process of enlarging our number systems comes to an end.
This makes $\mathbb{C}$ fundamentally different from the previous number systems in our chain. Because every nonconstant polynomial with coefficients in $\mathbb{C}$ has a solution in $\mathbb{C}$, we say that $\mathbb{C}$ is algebraically closed.
Note: The examples above motivate the successive enlargements; they do not show that each larger system is the smallest possible extension.

3 Historical motivation (Optional read)

The creation of complex numbers arose from attempts to solve cubic equations. If, in

$$ax^3 + bx^2 + cx + d = 0,$$

we make the substitution

$$x = y - \frac{b}{3a}$$

divide by a, and carry out the operations, an equation of the form

$$y^3 + py + q = 0$$

is obtained. At the beginning of the sixteenth century, the Italian algebraists Scipione del Ferro and Niccolò Tartaglia discovered a method for solving such equations. In modern notation it is equivalent to the formula

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{D}} + \sqrt[3]{-\frac{q}{2} - \sqrt{D}}, \qquad D = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3.$$

When the "discriminant" D is positive, this formula immediately gives a root of the equation. For example, for

$$y^3 = 9y + 28,$$

we have $D = 169$, and the formula gives $y = 4$, which is the unique real root.
It was discovered, however, that a phenomenon occurs here which is completely different from the case of quadratic equations. There are equations having real roots for which the discriminant D is negative. For example,

$$y^3 = 15y + 4$$

has the obvious root 4 (the other two roots are $-2+\sqrt{3}$ and $-2-\sqrt{3}$), but $D<0$. In this case the formula gives an expression involving square roots of negative numbers.
Thus mathematicians faced a dilemma: either abandon the del Ferro–Tartaglia method as a general method for solving cubic equations, or accept that an ordinary real number such as 4 might be represented by expressions containing square roots of negative numbers. The latter seemed inconceivable, but this did not prevent mathematicians from formally carrying out algebraic operations with such expressions. In the middle of the sixteenth century, Rafael Bombelli made bold assumptions about these expressions and showed that the apparently impossible calculation indeed produces the real root $y=4$. At the time, this discovery set up a dichotomy between real (in the sense of genuine) and imaginary (in the sense of non-existent) solutions to equations. The eighteenth-century mathematician Leonhard Euler introduced the symbol i for the number with $i^2=-1$.
Since then complex numbers have become an indispensable tool in mathematics and in its applications to the other sciences. Note that complex did not mean 'complicated'; it meant 'composed of several parts'. Jacques Hadamard, who in 1896 proved the Prime Number Theorem using complex analysis, wrote (echoing Painlevé) that the shortest path between two truths in the real domain often passes through the complex domain.

4 Construction of complex numbers

So far, writing $x+iy$ with $x,y\in\mathbb{R}$ and $i^2=-1$ is intuitive and we know it is useful to do operations with expressions of this form, but we have not yet rigorously constructed either i or the field of complex numbers.
In 1806 the French mathematician Jean-Robert Argand described a complex number $x+iy$ as a point (x,y) in the plane.

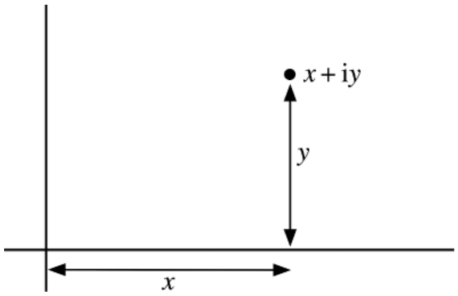


Figure 1: A complex number as a point in the plane

The horizontal axis became the real axis, the vertical axis the imaginary axis, and the number i was seen as the point one unit up on the imaginary axis.
We often describe Argand's representation of complex numbers as the 'complex plane', but as a set it is precisely the same as the 'real plane' $\mathbb{R}^2$.
Although the complex numbers were now described concretely as points in the plane, the mystification of earlier eras still shrouded them for most people. Gauss realised that the description

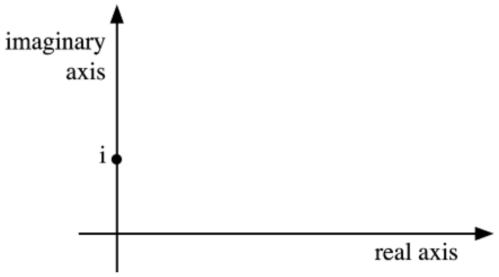


Figure 2: The real and imaginary axes

could be made even simpler: he clearly regarded a complex number as an ordered pair (x, y) of real numbers. This is the heart of the matter and the key to the modern description: a point in the plane is an ordered pair (x, y), and the symbol $x + iy$ is just another name for that point or that pair. The mysterious expression i is none other than the ordered pair $(0, 1)$.

Definition 4.1. The set of complex numbers is defined as

$$\mathbb{C} := \mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}.$$

We can define addition $+$ and multiplication $\cdot$ on $\mathbb{C}$ by

$$(x, y) + (u, v) = (x + u, y + v)$$
$$(x, y) \cdot (u, v) = (xu - yv, xv + yu)$$

$\forall (x, y), (u, v) \in \mathbb{C}$.

Theorem 4.1. $\mathbb{C}$ endowed with the operations described in Definition 4.1 is a field with $\mathbf{0} = (0, 0)$ and $\mathbf{1} = (1, 0)$.

Proof. **Exercise.** Part of the proof involves verifying that the additive and multiplicative inverses of (x, y) are $-(x, y) = (-x, -y)$ and, when $(x, y) \neq (0, 0)$, $(x, y)^{-1} = \left(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2}\right)$, respectively. □

Remark 1. If we defined "multiplication" by simply assigning $(x, y) \cdot (u, v) = (xu, yv)$, then we would have $(1, 0) \cdot (0, 1) = (0, 0) = \mathbf{0}$ while $(1, 0) \neq \mathbf{0}$ and $(0, 1) \neq \mathbf{0}$ which is impossible in a field.

We now note that $\forall (x, y) \in \mathbb{C}$ we can write $(x, y) = (x, 0) + (y, 0) \cdot (0, 1)$.

All the elements of the form $(x, 0)$ form a subset of $\mathbb{C}$ that can be put in a bijective correspondence with the set of real numbers $(x, 0) \mapsto x$. Moreover, this bijection preserves the arithmetic operations $(x, 0) + (u, 0) = (x + u, 0)$, $(x, 0) \cdot (u, 0) = (xu, 0)$.
From now on we will identify the subset $\{(x, 0) \in \mathbb{C}\} \subseteq \mathbb{C}$ with the set of real numbers. We next establish an important property of the special element $(0, 1)$.

Definition 4.2. Define $i = (0, 1)$.

Proposition 4.1. $i^2 = -1$, i.e. $(0, 1) \cdot (0, 1) = (-1, 0)$.

Proof. □

Remark 2. This construction gives us exactly what we wanted: a field containing a copy of $\mathbb{R}$ together with an element whose square is -1. Indeed, suppose F is any field containing $\mathbb{R}$ and an element j with $j^2 = -1$. Since F is closed under addition and multiplication, it must contain every element of the form $x + yj$ with $x, y \in \mathbb{R}$.

Definition 4.3. The **real part** of a complex number (x, y) is $\mathrm{Re}(x, y) = x$.
The **imaginary part** of a complex number (x, y) is $\mathrm{Im}(x, y) = y$.
We then write $(x, y) = x + iy$, and note that both real and imaginary parts of a complex number are real numbers.

Therefore from now on we will identify

$$(x, y) \quad \longleftrightarrow \quad x + iy.$$

Since two ordered pairs are equal exactly when their corresponding entries are equal, two complex numbers are equal if and only if they have the same real part and the same imaginary part:

$$x + iy = u + iv \iff x = u \quad \text{and } y = v \qquad (x, y, u, v \in \mathbb{R}).$$

Under this identification,

$$(x + iy)(u + iv) = xu + ixv + iyu + i^2yv.$$

Since $i^2 = -1$,

$$(x + iy)(u + iv) = (xu - yv) + i(xv + yu),$$

which corresponds exactly to

$$(x, y)(u, v) = (xu - yv, xv + yu).$$

It is natural to ask whether we can order $\mathbb{C}$.

Proposition 4.2. There is no order on $\mathbb{C}$ that makes it an ordered field.

Proof. Suppose, for a contradiction, that $<$ makes $\mathbb{C}$ an ordered field. Since $i \neq 0$ we must have $i^2 > 0$. Thus, $-1 > 0 \Rightarrow 1 < 0$, which is a contradiction since in every ordered field we must have $1 > 0$.
Note: You have shown that $x^2 > 0$ for $x \neq 0$ and that $1 > 0$ in an ordered field in HW4. □

Example 4.1. Consider $2+3i, 4+5i \in \mathbb{C}$. We then have

(i) $(2+3i)+(4+5i) = 6+8i$

(ii) $(2+3i)(4+5i) = 8+10i+12i+15i^2 = -7+22i$

(iii) $\dfrac{1}{2+3i} = (2,3)^{-1} = \left(\dfrac{2}{2^2+3^2}, \dfrac{-3}{2^2+3^2}\right) = \dfrac{2}{13} - \dfrac{3}{13}i$

5 Complex Conjugate and Modulus

We saw that the complex roots of the quadratic equation $ax^2+bx+c=0$ with negative discriminant were given by

$$\boxed{x = \frac{-b \pm i\sqrt{4ac-b^2}}{2a}}.$$

So, the two roots take the form

$$A+iB \qquad \text{and} \qquad A-iB.$$

where $A = \frac{-b}{2a}$ and $B = \frac{\sqrt{4ac-b^2}}{2a}$.
It holds more generally that if a polynomial has real coefficients and $A+iB$, $A, B \in \mathbb{R}$ is a root, then $A-iB$ is also a root.
We will call $A+iB$ and $A-iB$ **complex conjugates**.

Definition 5.1. The **complex conjugate** of $z = x+iy \in \mathbb{C}$ is

$$\overline{z} = x - iy \in \mathbb{C}.$$

Geometrically, $\overline{z}$ is the reflection of z across the real axis.

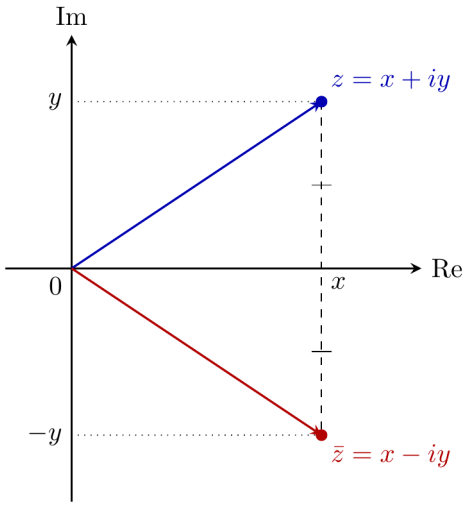


Figure 3: The complex conjugate: reflection across the real axis

Definition 5.2. The **modulus** (or the length, or the absolute value) of $z = x + iy \in \mathbb{C}$ is

$$|z| = \sqrt{x^2 + y^2} \in \mathbb{R}_{\geq 0}.$$

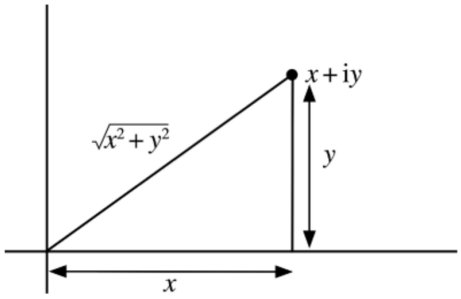


Figure 4: Modulus of a complex number: distance from the origin

Remark 3. If $r \in \mathbb{R}$, i.e. $\mathrm{Re}(r) = r$ and $\mathrm{Im}(r) = 0$, then the modulus of r coincides with the absolute value $|r| = \sqrt{r^2}$.

Remark 4. For $z = x + iy \neq 0$, we have

$$z\overline{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2 > 0.$$

Consequently,

$$\frac{1}{z} = \frac{\overline{z}}{x^2 + y^2} = \frac{\overline{z}}{|z|^2}.$$

Thus, to divide by a nonzero complex number, we multiply the numerator and denominator by its complex conjugate, making the denominator real and positive.

Examples

Write the following complex numbers in the form $x + iy$ with $x, y \in \mathbb{R}$.

(i) $\dfrac{4 - i}{3 + 2i} = \dfrac{(4 - i)(3 - 2i)}{(3 + 2i)(3 - 2i)} = \dfrac{10 - 11i}{13} = \dfrac{10}{13} - \dfrac{11}{13}i$

(ii) $\dfrac{1}{2 + 3i} + \dfrac{1 - i}{2 - 3i} = \dfrac{2 - 3i}{13} + \dfrac{(1 - i)(2 + 3i)}{13} = \dfrac{7}{13} - \dfrac{2}{13}i.$

Properties of the conjugate and the modulus

(i) $\overline{\overline{z}} = z$

(ii) $\overline{z+w} = \overline{z} + \overline{w}$

(iii) $z + \overline{z} = 2\,\mathrm{Re}(z)$ and $z - \overline{z} = 2i\,\mathrm{Im}(z)$

(iv) $\overline{z \cdot w} = \overline{z} \cdot \overline{w}$

(v) if $w \neq 0$ then $\overline{w} \neq 0$ and

$$\overline{\left(\frac{z}{w}\right)} = \frac{\overline{z}}{\overline{w}}$$

(vi) $z = \overline{z}$ if and only if $z \in \mathbb{R}$

(vii) $z = 0$ if and only if $\overline{z} = 0$

(viii) $|z| = |\overline{z}|$

(ix) $z \cdot \overline{z} = |z|^2$

(x) $|zw| = |z||w|$

(xi) $|\,\mathrm{Re}(z)| \leq |z|$ and $|\,\mathrm{Im}(z)| \leq |z|$

(xii) Triangle inequality: $|z + w| \leq |z| + |w|$

Proof. □